BENSON CYRIL NANA BOAKYE

Portfolio Website   +1 541-286-0930   nanaboab@gmail.com   LinkedIn   Github

SUMMARY

Results-driven Senior Data Analyst and Statistician with a Master of Science in Statistics and 4+ of progressive experience spanning data engineering, advanced analytics, machine learning, and enterprise-level reporting initiatives across complex, regulated environments. Expert in architecting and managing large-scale datasets across SQL Server, Oracle, and cloud environments, with strong proficiency in SQL, ETL pipeline development, and data governance.Proven ability to deliver executive-ready insights through scalable analytics solutions and interactive dashboards using Power BI and Tableau. Brings strong applied experience in statistical modeling, predictive analytics, experimental design, and clinical trial analysis using Python, R, SAS, and SQL, with a consistent focus on data integrity, regulatory compliance, and operational efficiency. Recognized for translating ambiguous business and research objectives into robust technical solutions, driving cross-functional collaboration, and delivering high-impact, data-driven outcomes.

CORE COMPETENCIES

- **Statistical Computing & Programming:** Python, R, SQL, SAS, Stan, MATLAB, Power BI, Tableau, Excel; advanced statistical programming, reproducible analysis, simulation, model development, data wrangling (pandas, dplyr, data.table), scripting automation, high-dimensional data handling, report generation, and workflow optimization.
- **Machine Learning & Advanced Analytics:** Predictive modeling, Supervised and Unsupervised learning, regression (linear, logistic, Poisson, survival), random forests, gradient boosting (XGBoost), neural networks, k-NN, PCA, and cross-validation.
- **Biostatistics & Clinical Research:** GLM, Bayesian inference, survival and multivariate analysis, hypothesis testing, experimental and observational study design, clinical trial analytics, longitudinal, public health data analysis and quality assurance.
- **Data Engineering & Governance:** ETL pipeline design, relational and cloud-based databases (SQL Server, Oracle, GCP), MongoDB, data validation, metadata management, audit checks, regulatory compliance, data integrity, and workflow optimization.
- **Leadership, Communication & Collaboration:** Executive reporting, client presentations, mentoring, cross-functional team leadership, translating complex analyses for technical and non-technical stakeholders, and driving data-informed decisions.

PROFESSIONAL EXPERIENCE

Statistical Consultant — **September 2024 – June 2025**
Oregon State University, Department of Statistics

- Optimized statistical designs and methodologies for 10+ biomedical studies, including randomized controlled trials and longitudinal studies, reducing required sample sizes by 20% while preserving statistical power and study integrity.
- Directed advanced genomic and transcriptomic analyses (RNA-seq) using R and Python, identifying clinically relevant biomarkers and contributing to peer-reviewed publications through rigorous statistical validation and reproducible workflows.
- Served as lead statistical advisor to multidisciplinary research teams, guiding the analysis of immune responses to tuberculosis antigens and their association with malnutrition using generalized linear models and logistic regression frameworks.
- Conducted end-to-end statistical analysis of respiratory and clinical biomarker data, evaluating relationships between breath condensate pH, reflux events, and tracheal aspirate pepsin levels using SAS to inform clinical interpretation.
- Designed and implemented predictive modeling pipelines using penalized logistic regression (LASSO) to model pollen fitness from high-dimensional genomic data, improving model (AUROC) by $\sim 70\%$ while enhancing interpretability and feature selection.
- Established reproducible statistical workflows in R, Python, and SAS, incorporating version control, validation checks, and documentation to support transparency, peer review, and publication readiness.

Graduate Teaching Assistant — **September 2023 – June 2025**
Oregon State University, Department of Statistics

- Led instruction on experimental design (randomization, blocking, and factorial designs), strengthening students' ability to design rigorous studies and apply data collection and analysis workflows, resulting in a measurable improvement in understanding.
- Integrated machine learning concepts such as support vector machines and decision trees into applied coursework using Python, developing scalable instructional materials that reduced preparation time by approximately 40%.
- Facilitated large-scale recitation sessions for 500+ students covering statistical methods and data analysis using Python and Excel, contributing to a 15% improvement in overall student performance and comprehension.
- Applied dimensionality reduction techniques including PCA and t-SNE in instructional settings to improve model interpretability and help students identify structure and patterns in high-dimensional data.
- Maintained academic integrity and data confidentiality while delivering timely, constructive feedback, fostering critical thinking, collaboration, and analytical rigor across diverse student cohorts.

Senior Inclusive Data & Research Analyst — **October 2022 – August 2023**
Laboratory for Interdisciplinary Statistical Analysis, KNUST

- Led the development of Bayesian hierarchical models to quantify healthcare access disparities, identifying a 30% gap across demographic groups by integrating informed priors from community-based survey data.
- Designed and executed survey methodologies using stratified sampling, post-stratification weighting, and multiple imputation, improving representation of underrepresented populations by 35% and reducing systematic bias in public health datasets.
- Served as a senior analytical contributor across 7+ interdisciplinary research projects examining the impacts of COVID-19 on healthcare disparities, vaccine uptake, and telehealth access, delivering statistical analyses and validated models using Python.
- Improved the representativeness of 10 public datasets by conducting stratified analyses and demographic weighting in R, identifying and correcting 20% undercoverage in marginalized groups, resulting in a 15% increase in data accuracy for equity-focused research.

Actuarial Analyst Intern **September 2021 – December 2021**
National Insurance Trust

- Applied unsupervised machine learning in python for anomaly detection on 2TB of data, reducing process time by 20%.
- Built predictive models in Python(Tensorflow) to improve policyholder behavior, improving risk profiling accuracy 24%.
- Created visual reports using Power BI and Tableau, applied run-off triangles and pricing models to claims, generated insights to ascertain competitiveness, and contributing to the preparation of Tables, Listings, and Figures (TLFs) for stakeholder review.
- Supported the application of Monte Carlo simulation methods for reinsurance optimization and asset-liability management analysis, contributing to investment strategies that achieved approximately 5% higher ROI for client portfolios.

Compensation & Benefits Analyst Intern **June 2019 – September 2019**
National Insurance Trust

- Supported a pay equity analysis project across 1,200+ employees, identifying gaps in base salary and incentives and providing actionable recommendations adopted by HR, resulting in a 10% increase in equitable adjustments..
- Performed variance analysis and data validation on salary, incentive, and benefits records, identifying discrepancies in $\sim 12\%$ of entries and contributing to corrections that reduced reporting errors by 20%.
- Developed and optimized SQL queries to extract, clean, and aggregate compensation and benefits data from multiple HR systems, reducing processing time by 25% and enabling faster reporting for leadership..
- Collaborated with finance and HR teams on vendor and payroll reconciliation, verifying 500+ monthly transactions, streamlining accounts payable processes, and reducing reporting errors by 15%.

RELEVANT PROJECTS

A Machine Learning Approach to Predicting Idiopathic Pulmonary Fibrosis Progression (IPF)

- Evaluated logistic regression, decision trees, random forests, LASSO, and ensemble methods on 1,129 proteomic features to predict IPF progression, with random forests achieving 83.3% accuracy and AUC of 0.984; implemented cross-validation and GridSearchCV hyperparameter tuning to ensure reproducibility and minimize overfitting, and performed feature importance ranking and SHAP analysis to identify key proteomic biomarkers for early intervention.

Survival Analysis of Treatment Efficacy in Primary Biliary Cirrhosis (PBC)

- Analyzed Cyclosporin A's effect on PBC survival using Kaplan-Meier, AFT, and Cox models, improving prediction accuracy by 15%; validated model assumptions, conducted residual and sensitivity analyses, and generated stratified survival curve visualizations to identify high bilirubin and low albumin as key risk factors, enhancing interpretability and guiding clinical decision-making.

Predictive Modeling and Data Analysis of Breast Cancer

- Designed and evaluated five classification models (Logistic Regression, KNN, Random Forest, SVM with RBF kernel, and Gradient Boosting) using FNA image features for breast cancer diagnosis, achieving up to 94% precision and 98.25% accuracy; applied feature scaling, PCA, and data augmentation to reduce dimensionality and improve model generalization, and evaluated trade-offs in sensitivity and specificity using confusion matrices, ROC-AUC, and F1-scores for clinical relevance.

Modeling Female Sex Worker Distribution in Sub-Saharan Africa for HIV Prevention

- Deployed mixed-effects models (negative binomial, zero-inflated, Poisson) to estimate FSW population counts, achieving AIC of 2863.208 and accounting for overdispersion and excess zeros; applied REML estimation to include country, region, and dataYear as random effects ($p < 0.0001$), capturing hierarchical variation and informing targeted HIV intervention planning. Performed diagnostic checks on residuals and model fit to improve reliability of resource allocation across multiple countries.

Air Quality Analysis with R-Shiny App

- Developed an interactive R-Shiny application to explore PM2.5 levels across cities in the USA and India, enabling dynamic trend analysis and regional comparisons; integrated event annotations such as COVID-19 lockdown periods to correlate policy interventions with air quality changes. Implemented statistical summaries, temporal smoothing, and visualization techniques to provide actionable insights for public health policy and environmental monitoring.

EDUCATION

Oregon State University **September 2023 – June 2025**
Master of Science in Statistics *CGPA: 3.74/4.0*

Kwame Nkrumah University of Science and Technology (KNUST) **August 2018 – September 2022**
Bachelor of Science in Actuarial Science *CGPA: 3.8/4.0*

PUBLICATION

- Fowotade, O.D., Makinde, J.O., Boakye, B.C.N., & Lasisi, S.O. (2025). Impact of Probiotics on Metabolic Interactions for the Prevention of Colorectal Cancer: A Comprehensive Network with Molecular Docking Studies. *Journal of Advances in Medical and Medical Research*, 37(6), 234–248. DOI link

CERTICATIONS & PROFESSIONAL MEMBERSHIP

- Member, American Statistical Association (ASA), Since 2023
- Getting Started with SAS Programming, SAS, Apr 2025
- Applied Data Analytics, WorldQuant University, 2025
- Data Science Foundations, Great Learning, Aug 2022
- Communicating with Emotional Intelligence, LinkedIn Learning, Dec 2024
- Strategic Thinking and Decision-Making, LinkedIn Learning, Dec 2024
- Conflict Resolution Foundations, LinkedIn Learning, Dec 2024